

Problem

Sketch the following functions:

(a) $x(t) = 10e^{-t}u(t-1)$,

(b) $y(t) = 10e^{-(t-1)}u(t)$,

(c) $z(t) = \cos 4t\delta(t-1)$

Step-by-step solution

Step 1 of 5

Write the expression for $x(t)$.

$$x(t) = 10e^{-t}u(t-1)$$

Define step function $u(t)$.

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

The unit step function $u(t-1)$ is defined as,

$$u(t-1) = \begin{cases} 1 & t \geq 1 \\ 0 & t < 1 \end{cases}$$

Calculate the value of $x(t)$ at $t = 1$ s.

$$\begin{aligned} x(t) &= 10e^{-(1)}u(1-1) \\ &= 3.68 \end{aligned}$$

Determine the value of $x(t)$ at $t = \infty$

$$\begin{aligned} x(t) &= 10e^{-(\infty)}u(\infty) \\ &= 0 \end{aligned}$$

Step 2 of 5

Sketch the function $x(t) = 10e^{-t}u(t-1)$.

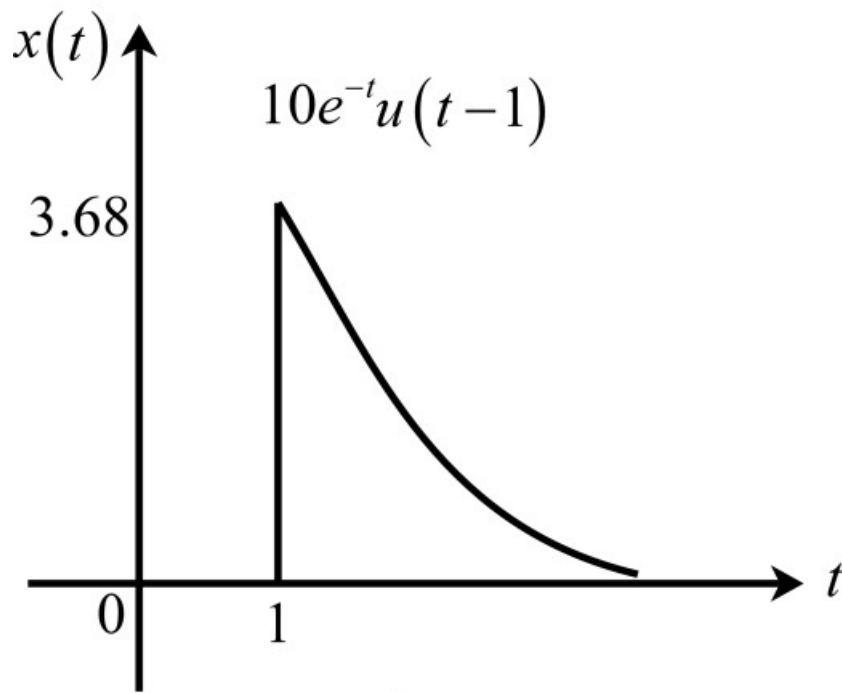


Figure 1

Write the expression for $y(t)$.

$$y(t) = 10e^{-(t-1)}u(t)$$

Define step function $u(t)$.

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

Calculate the value of $x(t)$ at $t = 0$ s.

$$\begin{aligned} y(t) &= 10e^{-(0-1)}u(0) \\ &= 27.18 \end{aligned}$$

Determine the value of $x(t)$ at $t = \infty$

$$\begin{aligned} y(t) &= 10e^{-(\infty)}u(\infty) \\ &= 0 \end{aligned}$$

Sketch the function $y(t) = 10e^{-(t-1)}u(t)$.

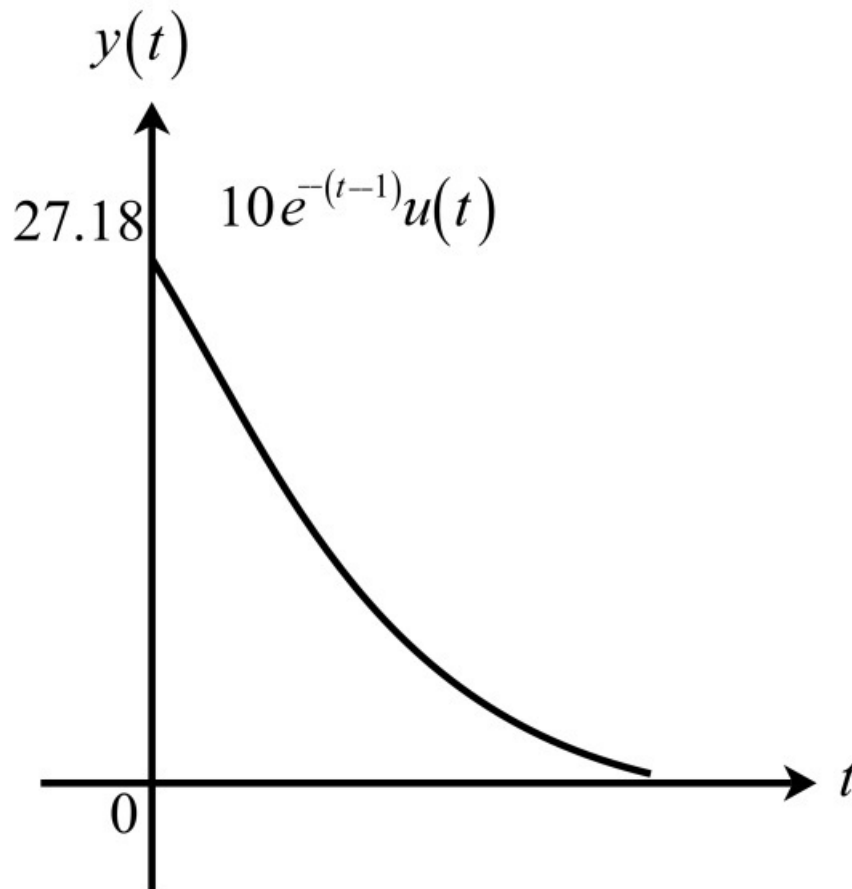


Figure 2

Step 4 of 5

Write the expression for $z(t)$.

$$z(t) = \cos 4t\delta(t-1)$$

Impulse function, $\delta(t)$ defines only at $t = 0$ s.

Unit delay impulse function, $\delta(t-1)$ defines only at $t = 1$ s.

Calculate the value of $z(t)$ at $t = 1$ s.

$$\begin{aligned} z(t) &= \cos 4(1)\delta(0) \\ &= -0.65 \end{aligned}$$

Sketch the function $z(t) = \cos 4t \delta(t-1)$.

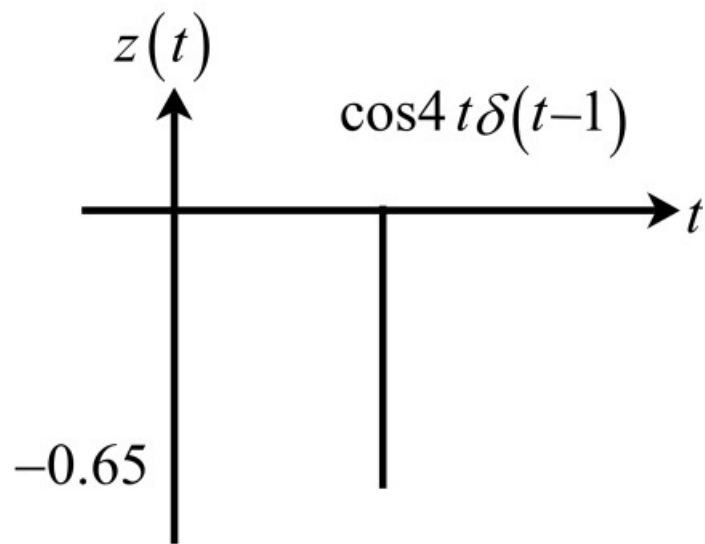


Figure 4